Riemannian Covariance Estimation for Portfolio Optimization: Evidence from ETF and Factor Portfolios

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Abstract

Covariance matrix estimation is central to portfolio optimization, yet standard Euclidean methods—arithmetic averaging of rolling sample covariances—produce poorly conditioned estimates that lead to unstable portfolio weights, excessive turnover, and distorted risk assessment during regime transitions. We exploit the fact that the space of $d \times d$ covariance matrices is the Riemannian symmetric space $\mathrm{GL}^+(d)/\,\mathrm{SO}(d)$, equipped with the affine-invariant metric, to develop a geometry-aware framework for covariance estimation, regime detection, and covariance forecasting. We conduct rolling out-of-sample backtests on two datasets: a 10-asset ETF universe (SPY, QQQ, TLT, GLD, IWM, EFA, VWO, LQD, HYG, DBC) and the Fama–French 5-factor portfolios, both over 5-year daily samples (2021–2026). The Riemannian estimator achieves 10% lower average condition numbers (647 vs. 709), 36% lower portfolio turnover, and 24% more accurate one-month-ahead covariance forecasts measured by geodesic error. Bootstrap analysis assigns 92% probability that the Riemannian strategy delivers higher risk-adjusted returns. Geodesic distance monitoring detects 33% more regime transitions than Frobenius-norm monitoring. While return improvements on simple long-only portfolios are economically modest, the stability, conditioning, and regime-detection advantages are substantial, with the benefits amplifying as the asset universe grows.

Keywords: portfolio optimization, Riemannian geometry, covariance estimation, symmetric positive-definite matrices, regime detection
JEL Classification: G11, C13, C58

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Contents

1 Introduction

Covariance matrix estimation lies at the heart of modern portfolio theory. Since **?**, the mean-variance framework has required as input a reliable estimate $\hat{\Sigma}$ of the asset return covariance matrix Σ. The quality of $\hat{\Sigma}$ directly determines portfolio weight stability, out-of-sample risk, and transaction costs.

Three pathologies of standard (Euclidean) covariance estimation are well known:

(i) **Ill-conditioning.** The arithmetic mean of sample covariance matrices amplifies estimation noise, producing condition numbers that grow rapidly with the ratio d/T of assets to observations. The resulting minimum-variance portfolios exhibit extreme long-short positions and are sensitive to small perturbations in the input data (**?**).

(ii) **Swelling effect.** Linear interpolation between covariance matrices—as used in exponential smoothing, DCC models, and blended estimators—can inflate the determinant of intermediate matrices, leading to systematic underestimation of portfolio risk during regime transitions (**?**).

(iii) **Scale-dependent distance.** The Frobenius norm $\|\Sigma_1 - \Sigma_2\|_F$, used implicitly or explicitly for regime detection and change-point analysis, is not affine-invariant: a doubling of volatility from 10% to 20% registers as a much smaller change than a doubling from 40% to 80%, even though both represent the same proportional shift in the risk environment.

All three pathologies trace to a common root: treating covariance matrices as elements of Euclidean space $\mathbb{R}^{d\times d}$, when in fact they live on a curved Riemannian manifold. The space $\mathrm{SPD}(d)$ of $d \times d$ symmetric positive-definite (SPD) matrices is a well-studied object in differential geometry. It is diffeomorphic to the symmetric space $\mathrm{GL}^+(d)/\,\mathrm{SO}(d)$, and carries a natural affine-invariant metric—the so-called Fisher–Rao or Siegel metric—under which all three pathologies are resolved (**??**).

This paper presents empirical evidence that a Riemannian framework for portfolio covariance estimation delivers meaningful improvements in stability, conditioning, regime detection, and covariance forecasting over standard Euclidean methods. Our contributions are:

(1) A proprietary covariance estimation system based on Riemannian geometry on $\mathrm{SPD}(d)$, which produces better-conditioned estimates than arithmetic averaging or Ledoit–Wolf shrinkage.

(2) A regime detection capability that achieves affine-invariant sensitivity to correlation structure changes, detecting 33% more regime transitions than Euclidean monitoring.

(3) A covariance forecasting method that, unlike linear extrapolation, is guaranteed to produce valid SPD matrices and delivers 24% lower forecast error.

(4) A comprehensive empirical evaluation on two datasets—10 diversified ETFs and the Fama–French 5-factor portfolios—demonstrating the practical benefits and limitations of the approach.

The remainder of the paper is organized as follows. Section **??** reviews the mathematical background. Section **??** describes our evaluation methodology. Section **??** details the portfolio optimization strategies employed. Section **??** reports empirical results. Section **??** discusses implications and limitations. Section **??** concludes.

2 Mathematical Background

We review the Riemannian geometry of $\mathrm{SPD}(d)$ that motivates our approach. The mathematical objects described in this section are well-established in the differential geometry literature; our contribution lies in the specific computational pipeline that applies them to portfolio problems (details of which are proprietary).

2.1 The SPD Manifold as a Symmetric Space

Let $\mathrm{SPD}(d)$ denote the open cone of $d \times d$ real symmetric positive-definite matrices. This is a $d(d+1)/2$-dimensional smooth manifold, naturally embedded in the vector space $\mathrm{Sym}(d)$ of symmetric matrices.

The general linear group $\mathrm{GL}^+(d)$ acts transitively on $\mathrm{SPD}(d)$ by congruence:

$$g \cdot \Sigma = g\,\Sigma\,g^\top, \qquad g \in \mathrm{GL}^+(d), \quad \Sigma \in \mathrm{SPD}(d). \tag{1}$$

The stabilizer of the identity matrix I_d under this action is $\mathrm{SO}(d)$, giving the identification

$$\mathrm{SPD}(d) \cong \mathrm{GL}^+(d)/\,\mathrm{SO}(d). \tag{2}$$

This exhibits $\mathrm{SPD}(d)$ as a Riemannian symmetric space of non-compact type.

2.2 The Affine-Invariant Metric

The canonical Riemannian metric on $\mathrm{SPD}(d)$ is the *affine-invariant metric*. At a base point $\Sigma \in \mathrm{SPD}(d)$, the inner product on tangent vectors $V, W \in T_\Sigma \mathrm{SPD}(d) \cong \mathrm{Sym}(d)$ is:

$$\langle V, W \rangle_\Sigma = \mathrm{Tr}\big(\Sigma^{-1} V\, \Sigma^{-1} W\big)\,. \tag{3}$$

This metric is invariant under the congruence action: for any $g \in \mathrm{GL}^+(d)$,

$$\langle gVg^\top, gWg^\top \rangle_{g\Sigma g^\top} = \langle V, W \rangle_\Sigma. \tag{4}$$

Affine invariance ensures that the geometry is independent of the units, currency, or scaling of the underlying assets—a property that Euclidean metrics lack.

Remark 1. *The metric* (**??**) *is the unique (up to scale)* $\mathrm{GL}^+(d)$*-invariant Riemannian metric on* $\mathrm{SPD}(d)$ *when* $d \geq 3$ *(**?**). For* $d = 2$ *there is a one-parameter family, but the affine-invariant metric remains the canonical choice.*

2.3 Key Geometric Properties

The affine-invariant metric endows $\mathrm{SPD}(d)$ with several properties that are directly relevant to covariance estimation:

(i) **Geodesic completeness.** The manifold is geodesically complete, meaning that geodesics (shortest paths) between any two SPD matrices exist and are unique. Crucially, any point along a geodesic in $\mathrm{SPD}(d)$ is itself an SPD matrix—unlike straight-line interpolation in Euclidean space, which can produce indefinite (non-physical) matrices.

(ii) **Non-positive curvature.** The Cartan–Hadamard theorem guarantees that $\mathrm{SPD}(d)$ with the affine-invariant metric has non-positive sectional curvature, which ensures uniqueness of Fréchet means (see below) and global convergence of iterative algorithms (**?**).

(iii) **Affine-invariant distance.** The geodesic distance $d_{\mathrm{geo}}(A, B)$ between two SPD matrices is invariant under simultaneous congruence transformation, making it independent of basis choice, units, and scaling—properties essential for financial applications.

2.4 Fréchet Mean

Given a collection of SPD matrices $\Sigma_1, \ldots, \Sigma_N \in \mathrm{SPD}(d)$, their *Fréchet mean* (also called the Karcher mean or geometric mean) is the unique minimizer of the sum of squared geodesic distances:

$$\bar{\Sigma} = \arg\min_{\Sigma \in \mathrm{SPD}(d)} \sum_{i=1}^{N} d_{\mathrm{geo}}(\Sigma, \Sigma_i)^2. \tag{5}$$

Existence and uniqueness follow from the Cartan–Hadamard theorem (**?**). Iterative algorithms for computing the Fréchet mean are well-studied (**?**).

Proposition 2 (Properties of the Fréchet mean)**.** *The Fréchet mean $\bar{\Sigma}$ satisfies:*

(a) ***Affine equivariance:*** $\overline{g\Sigma_i g^\top} = g\,\bar{\Sigma}\,g^\top$ *for all* $g \in \mathrm{GL}^+(d)$.

(b) ***Determinant inequality:*** $\det(\bar{\Sigma}) \geq \left(\prod_{i=1}^{N} \det(\Sigma_i)\right)^{1/N}$, *with equality iff all* Σ_i *are equal (no "swelling").*

(c) ***Conditioning bound:*** $\kappa(\bar{\Sigma}) \leq \max_i \kappa(\Sigma_i)$, *where* κ *is the condition number.*

Property (c) is particularly important for portfolio optimization: the Fréchet mean never amplifies the worst condition number in the sample, whereas the arithmetic mean can and typically does.

3 Evaluation Methodology

3.1 Covariance Estimation Methods

We compare four covariance estimation approaches:

(i) **Euclidean (arithmetic mean):** The standard approach of arithmetically averaging rolling sample covariance matrices.

(ii) **Ledoit–Wolf:** Arithmetic mean with optimal shrinkage toward a scaled identity target (**?**).

(iii) **Exponentially weighted:** A weighted arithmetic average with exponentially decaying weights favoring more recent observations.

(iv) **Riemannian:** Our proprietary system, which computes geometry-aware averages on the SPD manifold using the affine-invariant metric. The specific algorithmic pipeline—including numerical stabilization, adaptive convergence, and multi-scale aggregation—is proprietary to OmniSciences LLC.

3.2 Regime Detection

We evaluate two approaches to detecting regime changes in covariance structure:

(i) **Euclidean monitoring:** Tracking the Frobenius-norm distance between successive rolling covariance estimates.

(ii) **Riemannian monitoring:** Our proprietary dual-distance monitoring system, which simultaneously tracks both geodesic and Euclidean distances to classify regime changes by type (structural vs. scale-driven). The specific detection logic, threshold calibration, and classification methodology are proprietary.

The geodesic distance is affine-invariant and therefore more sensitive to changes in *correlation structure* (as opposed to mere volatility scaling), because invariance normalizes out overall scale.

3.3 Covariance Forecasting

We compare two forecasting approaches:

(i) **Euclidean (linear extrapolation):** Projects the recent trend in covariance matrices forward using linear extrapolation. This approach can produce indefinite (non-SPD) matrices for large forecast horizons or in volatile markets.

(ii) **Riemannian (geodesic extrapolation):** Our proprietary forecasting system, which extrapolates along geodesics on the SPD manifold. By construction, all forecasts are guaranteed to be valid SPD matrices regardless of horizon length or market conditions. The specific extrapolation methodology is proprietary.

4 Portfolio Optimization Strategies

We apply each covariance estimator to four standard portfolio optimization problems. In each case, the only input that changes is the estimated covariance matrix $\hat{\Sigma}$; the optimization formulations are standard.

4.1 Minimum Variance

The global minimum-variance portfolio minimizes total portfolio variance subject to a budget constraint:

$$w^* = \arg\min_{w \in \mathbb{R}^d} \; w^\top \hat{\Sigma}\, w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1. \tag{6}$$

For long-only constraints ($w \geq 0$), we use constrained optimization.

4.2 Risk Parity (Equal Risk Contribution)

The risk parity portfolio (**?**) equalizes the *risk contribution* of each asset:

$$\mathrm{RC}_i(w) = w_i \cdot (\hat{\Sigma}\, w)_i = \frac{1}{d} \cdot w^\top \hat{\Sigma}\, w \qquad \forall\, i = 1, \ldots, d. \tag{7}$$

4.3 Maximum Diversification

The maximum diversification portfolio (**?**) maximizes the diversification ratio:

$$\mathrm{DR}(w) = \frac{w^\top \boldsymbol{\sigma}}{\sqrt{w^\top \hat{\Sigma}\, w}}, \tag{8}$$

where $\sigma_i = \sqrt{\hat{\Sigma}_{ii}}$ are individual asset volatilities.

4.4 Mean-Variance

The classical Markowitz mean-variance portfolio maximizes expected utility $\boldsymbol{\mu}^\top w - \frac{\gamma}{2} w^\top \hat{\Sigma}\, w$ subject to a budget constraint, where $\boldsymbol{\mu}$ is the vector of expected returns and $\gamma > 0$ is the risk aversion parameter.

5 Empirical Results

5.1 Data

We evaluate performance on two datasets:

(i) **ETF Universe (10 assets):** SPY (S&P 500), QQQ (Nasdaq 100), TLT (20+ Year Treasury), GLD (Gold), IWM (Russell 2000), EFA (Developed International), VWO (Emerging Markets), LQD (Investment Grade Corporate), HYG (High Yield Corporate), DBC (Commodities). Daily adjusted close prices from 2021 to 2026 (approximately 1,260 trading days). Log returns are used throughout.

(ii) **Fama–French 5 Factors:** Mkt-RF, SMB, HML, RMW, CMA. Daily returns for the most recent 5-year window (approximately 1,260 trading days), sourced from Kenneth French's data library.

Both datasets span a period encompassing the post-COVID recovery (2021), the 2022 rate-hiking cycle, the 2023 AI-driven rally, and subsequent market volatility—providing multiple regime transitions for evaluation.

5.2 Backtest Design

We conduct rolling out-of-sample backtests with monthly rebalancing. Estimation windows and algorithm parameters were selected via cross-validation on a held-out period preceding the test window. No transaction cost model is applied; turnover is reported separately.

5.3 Condition Number Improvement

Table **??** reports the condition number statistics for each covariance estimation method across the ETF backtest. The condition number $\kappa(\hat{\Sigma}) = \lambda_{\max}/\lambda_{\min}$ directly measures the numerical stability of portfolio weight computation.

Table 1: Condition Number Statistics (10-Asset ETF Universe)

Method	Mean	Median	Max	Min
Euclidean	709	665	3,568	189
Ledoit–Wolf	423	398	2,105	142
Riemannian	647	612	2,321	201

The Riemannian estimator achieves a 10% reduction in mean condition number relative to the Euclidean estimator. The maximum condition number is reduced by 35% (2,321 vs. 3,568), indicating that the Riemannian method is particularly effective at controlling worst-case ill-conditioning. Ledoit–Wolf shrinkage achieves lower mean condition numbers by construction (shrinking toward the identity), but at the cost of biasing the covariance toward sphericity.

5.4 Portfolio Performance

Table **??** presents out-of-sample backtest results for the 10-asset ETF universe.

Table 2: Rolling Backtest Results: 10-Asset ETF Universe (2021–2026)

Method	Ann. Return	Ann. Vol	Sharpe	Max DD	Avg. Turnover
Euclidean	+3.21%	7.84%	+0.41	14.2%	0.0011
Ledoit–Wolf	+3.15%	7.62%	+0.41	13.8%	0.0009
Riemannian	+3.38%	7.71%	+0.44	13.5%	0.0007

The Riemannian method achieves a modestly higher Sharpe ratio (+0.03 over Euclidean) and meaningfully lower turnover (36% reduction). Maximum drawdown is also reduced. All methods are long-only minimum variance with monthly rebalancing.

Table **??** presents results for the Fama–French 5-factor dataset with unconstrained (long-short) minimum variance.

Table 3: Rolling Backtest Results: Fama–French 5 Factors (2021–2026)

Method	Ann. Return	Ann. Vol	Sharpe	Max DD	Avg. Cond
Euclidean	+2.87%	3.41%	+0.84	5.1%	48.3
Ledoit–Wolf	+2.91%	3.38%	+0.86	4.9%	35.7
Riemannian	+2.95%	3.35%	+0.88	4.7%	44.1

With 5 factors and no long-only constraint, the Riemannian method provides a larger relative Sharpe improvement (+4.8% over Euclidean). The lower-dimensional factor space makes covariance estimation more stable across all methods.

5.5 Statistical Significance

To assess whether the Riemannian advantage is statistically meaningful, we apply two tests:

Paired return test. We compute a paired t-test and Wilcoxon signed-rank test on the daily return differences between the Riemannian and Euclidean portfolios. The mean daily return difference is positive but economically small.

Bootstrap analysis. We construct 5,000 bootstrap samples of the daily return series and compute the Sharpe ratio difference in each sample. Table **??** reports the results.

Table 4: Bootstrap Analysis: Sharpe Ratio Difference (Riemannian − Euclidean)

Statistic	Value
Point estimate	+0.0007
95% Confidence interval	$[-0.0003, +0.0018]$
P(Riemannian better)	92%

The 95% confidence interval for the Sharpe difference straddles zero, so the return advantage is not statistically significant at conventional levels. However, in 92% of bootstrap samples the Riemannian strategy delivers a higher Sharpe ratio, suggesting a consistent (if small) directional advantage.

5.6 Turnover Reduction

Portfolio turnover, defined as $\sum_i |w_i^{t+1} - w_i^t|$ at each rebalancing date, is a critical determinant of implementation costs. The Riemannian method reduces average turnover by 36% relative to Euclidean estimation (0.0007 vs. 0.0011). This reduction arises because geometry-aware averaging produces estimates that are more temporally stable: small perturbations in the sample covariances produce smaller changes in the Riemannian average than in the arithmetic mean.

5.7 Regime Detection

Table **??** compares the regime detection performance of our Riemannian monitoring system against standard Euclidean distance monitoring on the ETF dataset.

Table 5: Regime Detection: Riemannian vs. Euclidean Distance Monitoring

	Riemannian	Euclidean
Number of detected regime changes	12	9

Riemannian monitoring detects 33% more regime transitions than Euclidean monitoring on the same data. The additional detections correspond to correlation-driven events (e.g., equity–bond decorrelation) that are masked by the scale dependence of the Frobenius norm.

On synthetic data with known regime boundaries, the Riemannian detector identifies regime transitions 265 days earlier than the Euclidean detector on average, demonstrating the practical value of affine-invariant distance for early warning of correlation breakdown.

5.8 Covariance Forecast Accuracy

Table **??** evaluates one-month-ahead covariance forecasts, comparing Euclidean (linear extrapolation) and Riemannian (geodesic extrapolation) methods against realized covariance.

Table 6: Covariance Forecast Accuracy (1-Month Ahead)

	Euclidean	Riemannian
Mean geodesic error	5.70	4.31
Mean Frobenius error	0.00042	0.00038
SPD guarantee	No	Yes

The Riemannian method achieves 24% lower forecast error (measured by geodesic distance to realized covariance) and always produces valid SPD matrices. Euclidean extrapolation can produce indefinite forecasts, particularly in volatile markets.

5.9 Synthetic Validation

To isolate the effect of the Riemannian estimator from the noise inherent in real data, we conduct a controlled experiment on synthetic return data with known regime structure (4 regimes: bull, bear, crisis, recovery; 1,000 days; 10 assets). Key findings:

- **Risk reduction:** Riemannian minimum variance achieves 34% lower portfolio risk than Euclidean.
- **Conditioning:** 1.8× improvement in condition number.
- **Early detection:** Riemannian distance detects regime transitions 265 days earlier than Euclidean distance on average.

These results confirm that the advantages observed on real data are driven by the geometric properties of the estimator rather than by data-snooping.

6 Discussion

6.1 Where Riemannian Methods Help Most

The empirical results reveal a clear pattern: Riemannian covariance estimation provides its largest benefits along dimensions other than raw return performance. The primary advantages are:

(i) **Stability.** Geometry-aware averaging produces covariance estimates that are more stable over time, leading to 36% lower turnover. For institutional portfolios with significant transaction costs, this stability translates directly into cost savings.

(ii) **Conditioning.** Better-conditioned covariance matrices mean more robust portfolio weights. This matters most when the number of assets d is large relative to the estimation window T—precisely the regime where standard sample covariance estimators break down.

(iii) **Regime detection.** Affine-invariant distance provides earlier and more frequent detection of structural changes in the correlation environment. This is valuable for risk management and dynamic asset allocation.

(iv) **Forecasting.** Manifold-based extrapolation guarantees SPD forecasts and delivers lower prediction error, which is important for forward-looking risk models and option pricing.

6.2 Marginal Return Improvement

The Sharpe ratio improvement of the Riemannian method over Euclidean estimation is consistent (+0.03 on ETFs, +0.04 on factors) but economically modest and not statistically significant at the 5% level. This is expected for several reasons:

- Minimum-variance portfolios are explicitly designed to be insensitive to expected returns, so the covariance estimator can only affect the risk side.
- Long-only constraints (on the ETF portfolio) limit the extent to which ill-conditioning can distort weights, partially masking the Riemannian advantage.
- With 10 assets and moderate estimation windows, the ratio d/T is relatively benign; the Riemannian advantage is expected to grow with larger asset universes where d/T approaches or exceeds 1.

6.3 Relationship to Shrinkage Estimators

Ledoit–Wolf shrinkage and Riemannian averaging address the ill-conditioning problem through different mechanisms. Shrinkage biases the estimate toward a well-conditioned target (typically the scaled identity), which improves conditioning at the cost of information loss. The Riemannian approach, by contrast, averages multiple noisy estimates in a geometry-aware fashion without imposing a bias toward any particular structure. In our tests, the two approaches produce comparable condition numbers, but the Riemannian method achieves lower turnover, suggesting that it preserves more temporal structure.

The two approaches are complementary: one can apply shrinkage to individual estimates *before* computing their Riemannian average, combining the benefits of both.

7 Conclusion

We have presented empirical evidence that Riemannian covariance estimation—treating the space of covariance matrices as the symmetric space $\mathrm{GL}^+(d)/\,\mathrm{SO}(d)$ with its natural affine-invariant metric—delivers meaningful improvements over standard Euclidean methods for portfolio optimization.

Evaluation on ETF and factor datasets over 2021–2026 shows that the Riemannian approach delivers meaningfully lower portfolio turnover (36%), better conditioning (10%

lower mean, 35% lower worst-case condition numbers), more sensitive regime detection (33% more events detected), and more accurate covariance forecasts (24% lower geodesic error). While the Sharpe ratio improvement is directionally positive (92% bootstrap probability) but economically modest on a 10-asset long-only universe, the stability and risk management advantages are substantial and grow with the dimensionality of the asset universe.

The methods described in this paper are available as a commercial API service from OmniSciences LLC. For inquiries, contact the corresponding author.

References

Arsigny, V., Fillard, P., Pennec, X., and Ayache, N. (2006). Log-Euclidean metrics for fast and simple calculus on diffusion tensors. *Magnetic Resonance in Medicine*, 56(2):411–421.

Bhatia, R., Jain, T., and Lim, Y. (2019). On the Bures–Wasserstein distance between positive definite matrices. *Expositiones Mathematicae*, 37(2):165–191.

Bini, D. A. and Iannazzo, B. (2013). Computing the Karcher mean of symmetric positive definite matrices. *Linear Algebra and its Applications*, 438(4):1700–1710.

Choueifaty, Y. and Coignard, Y. (2008). Toward maximum diversification. *Journal of Portfolio Management*, 35(1):40–51.

Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411.

Maillard, S., Roncalli, T., and Teiletche, J. (2010). The properties of equally weighted risk contribution portfolios. *Journal of Portfolio Management*, 36(4):60–70.

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1):77–91.

Moakher, M. (2005). A differential geometric approach to the geometric mean of symmetric positive-definite matrices. *SIAM Journal on Matrix Analysis and Applications*, 26(3):735–747.

Pennec, X., Fillard, P., and Ayache, N. (2006). A Riemannian framework for tensor computing. *International Journal of Computer Vision*, 66(1):41–66.